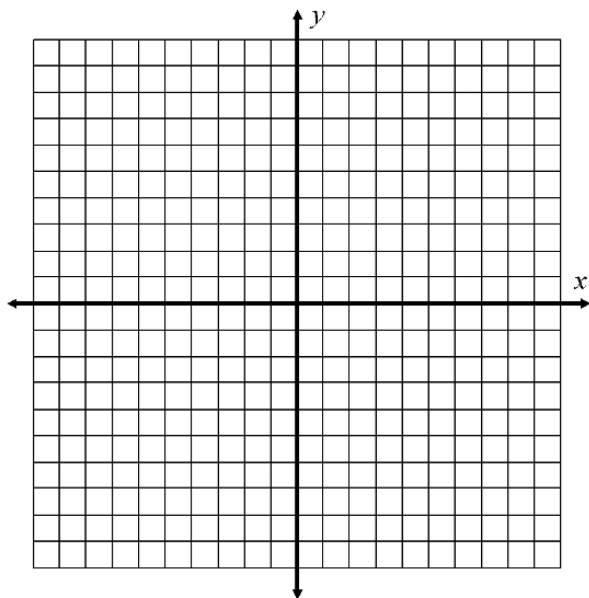
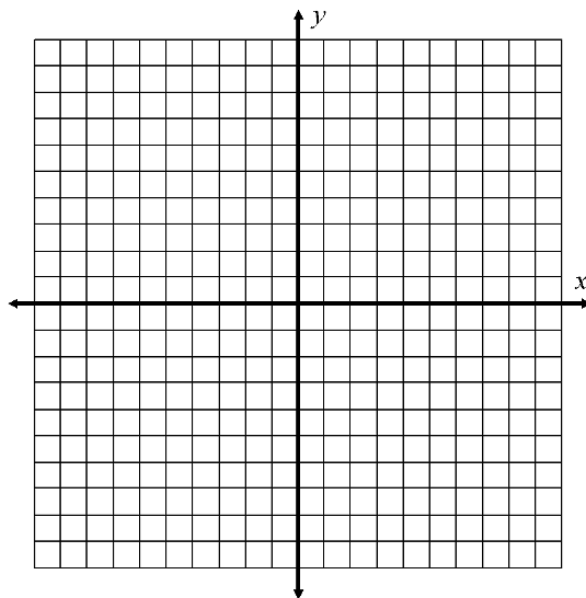


Assignment 2.03: Rotations

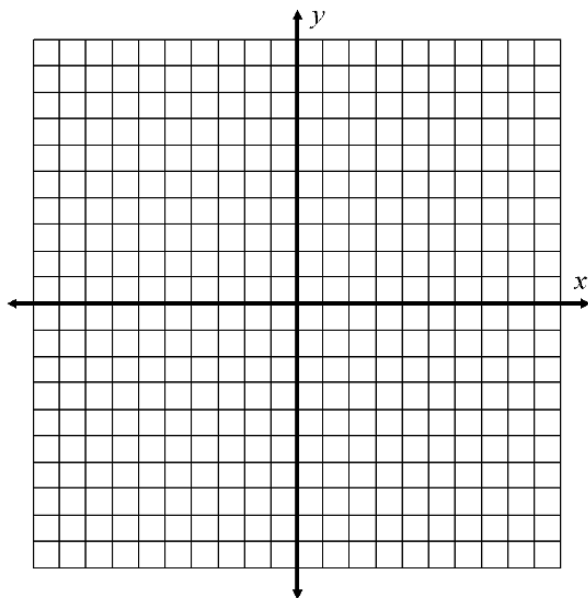
1. Triangle ABC has vertices $A(-2,8)$, $B(-1,1)$, and $C(-4,3)$. Find the coordinates of the vertices of $\triangle A'B'C'$ after a rotation of 90° counterclockwise about the origin.



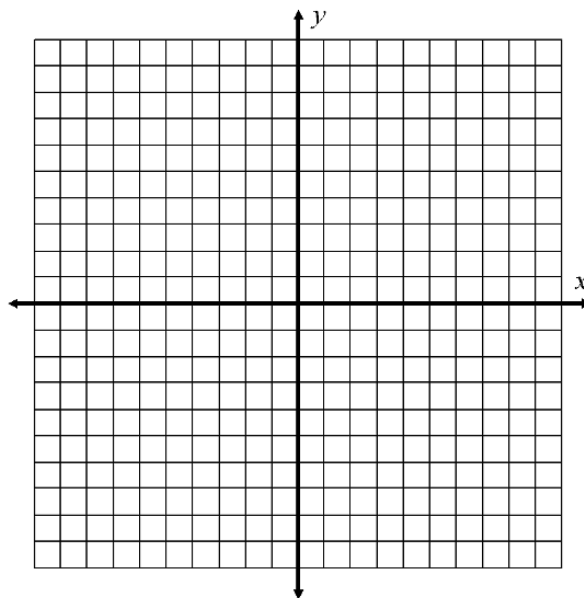
2. Triangle DEF has vertices $D(-7,1)$, $E(-6,5)$, and $F(-1,4)$. Find the coordinates of the vertices of $\triangle D'E'F'$ after a rotation of 270° counterclockwise about the origin.



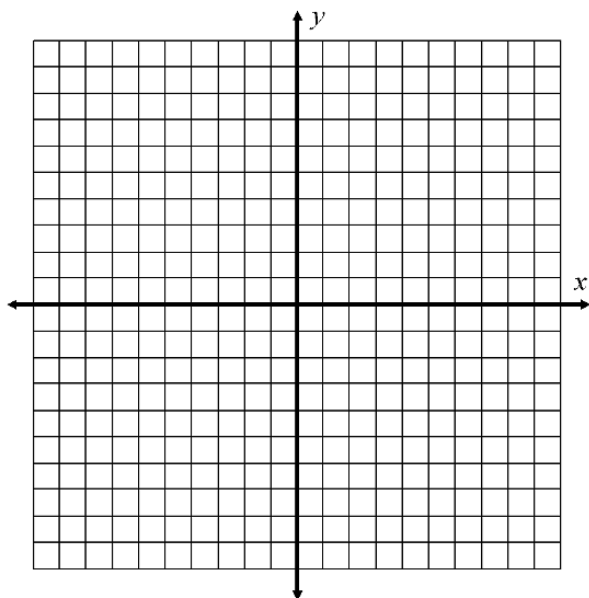
3. Triangle GHI has vertices $G(0,0)$, $H(-8,5)$, and $I(-5,7)$. Find the coordinates of the vertices of $\triangle G'H'I'$ after a rotation of 180° counterclockwise about the origin.



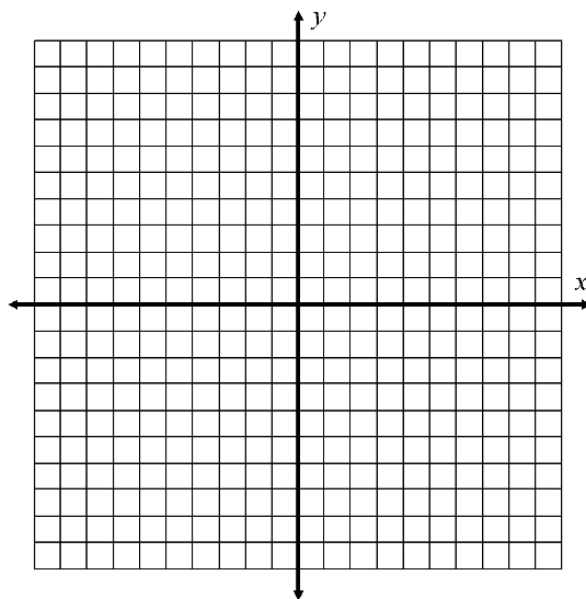
4. Triangle PQR has vertices $P(8,-2)$, $Q(2,-7)$, and $R(1,-4)$. Find the coordinates of the vertices of $\triangle P'Q'R'$ after a rotation of 540° counterclockwise about the origin.



5. Triangle ABC has vertices $A(-8,1)$, $B(-3,5)$, and $C(-1,2)$. Find the coordinates of the vertices of $\triangle A'B'C'$ after **reflection** across the point $(0,0)$.



6. Triangle DEF has vertices $D(3,-1)$, $E(5,-7)$, and $F(3,-1)$. Find the coordinates of the vertices of $\triangle D'E'F'$ after rotation of 90° **clockwise** about the origin.



7. Fill in the missing information.

$R_{90^\circ}(4, 7) = (\quad , \quad)$	$R_{90^\circ}(-2, -8) = (\quad , \quad)$	$R_{90^\circ}(9, -5) = (\quad , \quad)$
$R_{180^\circ}(4, 7) = (\quad , \quad)$	$R_{180^\circ}(-2, -8) = (\quad , \quad)$	$R_{180^\circ}(9, -5) = (\quad , \quad)$
$R_{270^\circ}(4, 7) = (\quad , \quad)$	$R_{270^\circ}(-2, -8) = (\quad , \quad)$	$R_{270^\circ}(9, -5) = (\quad , \quad)$